**model & grid** §3.1, 3.2

**training** §3.3

**inference** §3.4

**prior (grid)**

$\theta_2$

Sobol samples

$p(\theta)$ $\theta_1$

$\theta_i$

**dataset**

$\{\theta_i, x_i\}$

train

$\theta$

**RADMC-3D (simulator)**

$F_\nu$ $\lambda$

$\mathcal{G}(\theta; \lambda)$

$x_i$

**neural emulator (surrogate)**

§3.3.2 & Fig. B.1

$\theta$ $\theta$ $F_\nu$

$\widehat{\mathcal{G}}_\phi(\theta; \lambda)$

$\widehat{\mathcal{G}}_\phi$

**prior (target)**

$\theta_2$ $p(\theta)$ $\theta_1$

$p(\theta)$

$\widehat{\mathcal{G}}_\phi$

**prior predictive**

$F_\nu$ $\lambda$ $p(x)$

**nested sampling**

$\theta_2$ $\theta_1$

$x_{\text{obs}}$

**observation (target)**

$F_\nu$ $\lambda$ $x_{\text{obs}}$

$p(\theta \mid x_{\text{obs}})$

**posterior (target)**

$\theta_2$ $\theta^\star$ $p(\theta \mid x_{\text{obs}})$ $\theta_1$

**validation** §4

$\mathcal{G}$

**RADMC-3D prediction**

$F_\nu$ $\lambda$ $\mathcal{G}(\hat{\theta}; \lambda)$

compare

**neural emulator prediction**

$F_\nu$ $\lambda$ $\widehat{\mathcal{G}}_\phi(\hat{\theta}; \lambda)$

$\widehat{\mathcal{G}}_\phi$

$\hat{\theta}$, $\theta^\star$

$\hat{\theta}$